CAPSTONE · APPLIED AGENTIC AI FOR PMS/TPMS

Mira

An AI project intelligence assistant for Nexora's PMs and TPMs

Multi-agent n8n workflow · OpenAI gpt-4o-mini · LangSmith tracing and evaluation

13/13
baseline tests passed

100%
groundedness

$0.0008
per project plan

THE PROBLEM

Nexora's PMs spend their week assembling documents, not running projects

3–4 h

to turn a brief into a project plan

1–2 h

every week per status report

Ad hoc

risk matrices: detailed on some projects, missing on others

Too late

missed milestones surface only in retrospectives

Many concurrent projects, a small team, and no standard format for plans, risks or status reports.

Source: Mira problem statement, Nexora Pvt. Ltd.

WHAT MIRA DOES

Five capabilities, one chat window

CORE

Project plan

8 phases, milestones and metrics, each item quoted from the brief

CORE

Risk assessment

All 10 register risks rated, ranked and linked to live task signals

CORE

Weekly status

Counts, RAG, blockers and focus for any sprint or the whole project

EXTENDED

Milestone alerts

At-risk milestones, blocked and overdue tasks, escalations

EXTENDED

Stakeholder email

Sprint-scoped update with exact task names, honest about slippage

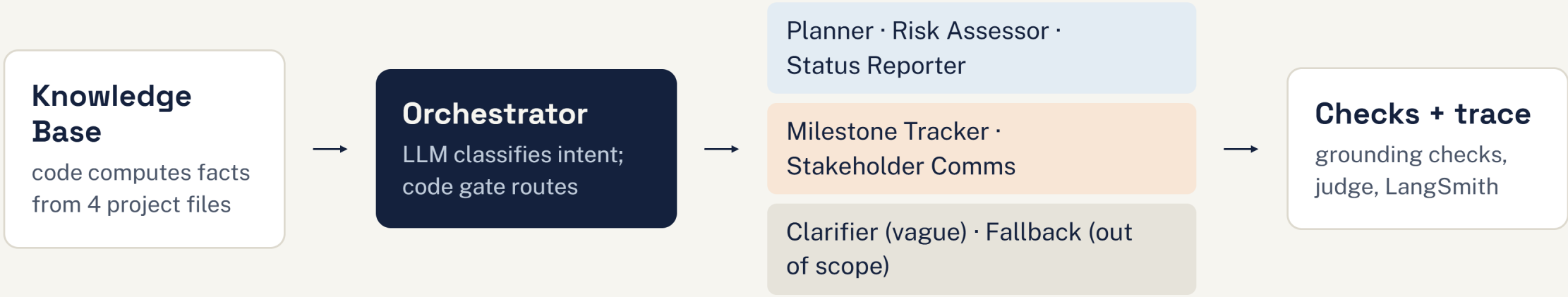
GUARDRAIL

Clarifier

Vague request? Mira asks for scope, goals, timeline and team instead of inventing

ARCHITECTURE

Router pattern: one specialist per request



Why a router

Requests vary. One LLM call instead of five: cheaper, faster, each agent testable alone.

Three nodes per agent

Build Prompt (code) → OpenAI with strict JSON schema → Format Output + checks (code).

Stack

n8n Cloud · gpt-4o-mini (gpt-4o judge) · LangSmith via REST · data embedded.

DESIGN PRINCIPLE

Code computes. The LLM only writes.

Facts in code

Counts, sprints, overdue and blocked flags, milestone links, risk ranking and every table

Schemas restrict output

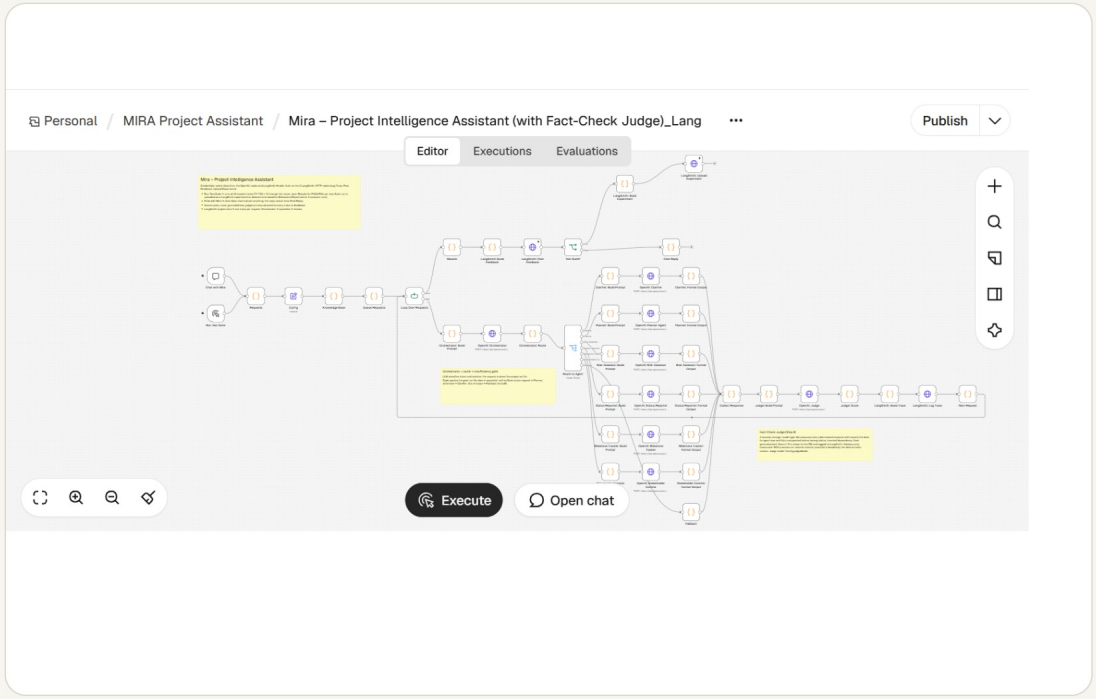
Enums allow only in-scope task IDs, all 10 risk IDs, and verbatim quotes from the brief

Checks on every answer

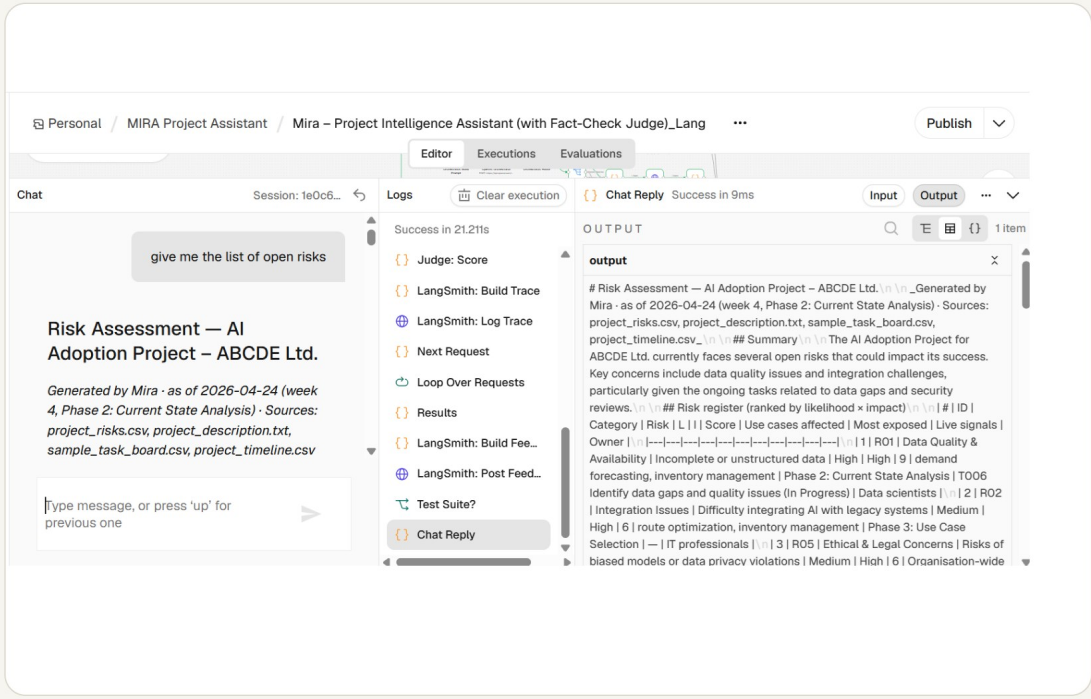
Quotes, IDs, numbers, dates and milestone names verified; score shown to the PM

MIRA IN ACTION

The n8n workflow and a live chat answer



47-node workflow: chat and test-suite triggers, router, 6 agents, judge, LangSmith



“Give me the list of open risks”, routed to the Risk Assessor, answered in 21 s

BASELINE RESULTS

All 12 baseline tests pass

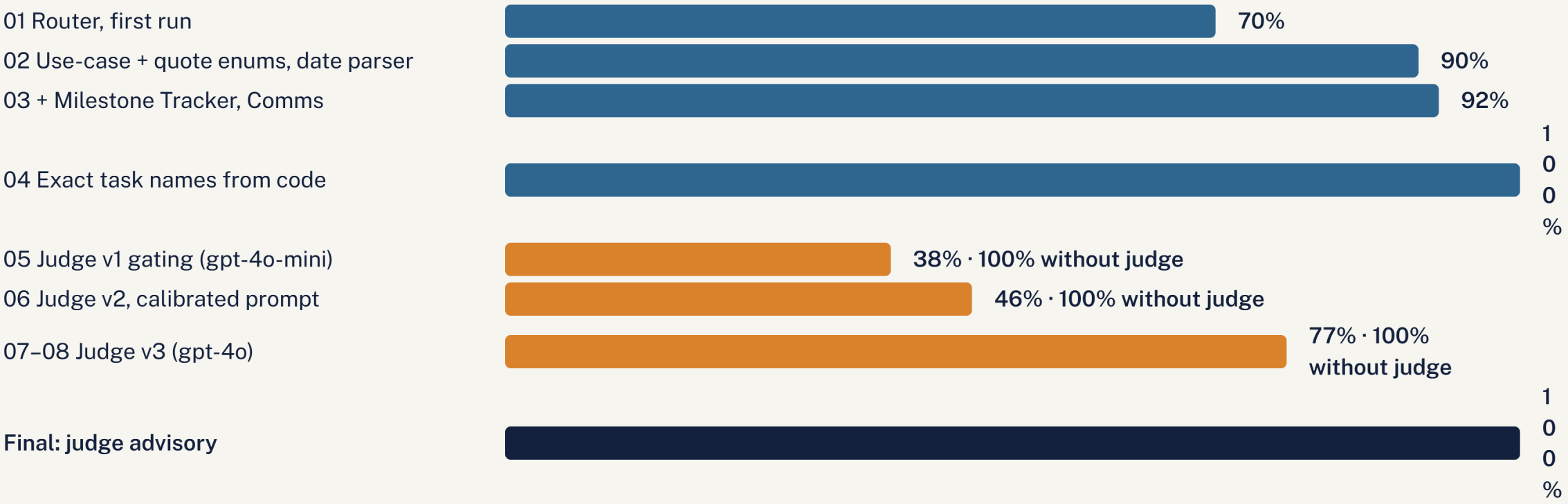
13/13

ID	Request	Route	Verified	Result
T1	Project plan, ABCDE	planner	8 phases; 19/19 items quoted from brief	PASS
T2 · T9	Vague plan requests	clarifier	Asks scope, goals, timeline, team; no plan	PASS
T3 · T7	Risk assessment · top 3 risks	risk	All 10 register risks rated; none invented	PASS
T4 · T6	Vague risk / status	clarifier	States missing data; no risks, no %	PASS
T5	Status, Sprint 3	status	Counts 0/2/0/2; real task names	PASS
T8	Blocked / at-risk tasks	milestone	T024 Security review blocked; T006 overdue	PASS
T10	Status counts, whole project	status	Done 5 · In Progress 3 · Blocked 1 · To Do 16	PASS
T11	Milestones, next 2 weeks	milestone	Audit report + gap analysis report (04-26)	PASS
T12	Stakeholder email, Sprint 2	comms	Only Sprint 2 tasks; professional	PASS
X1	“Weather in Paris?”	fallback	Out-of-scope reply, no LLM narrative	PASS

100% groundedness · 100% routing accuracy · gpt-4o-mini · as-of 2026-04-24 · full results in results/07_mira_complete_2026-11-02.json

IMPROVEMENT HISTORY

From 70% to 100%, one constraint at a time



Blue: deterministic checks only. Orange: runs where the LLM judge also gated the test; every one of those failures was a judge flag, not a Mira error.

Pass rate per run · 10 experiments on LangSmith dataset mira-baseline

EVALUATION INSIGHT

The LLM judge needed judging too

≈ 9%

Judge v1 · gpt-4o-mini

precision on 34 flagged claims; flagged hedged risk language as errors

≈ 0%

Judge v2 · calibrated

prompt rules and examples did not fix it: a model capability limit

33–67%

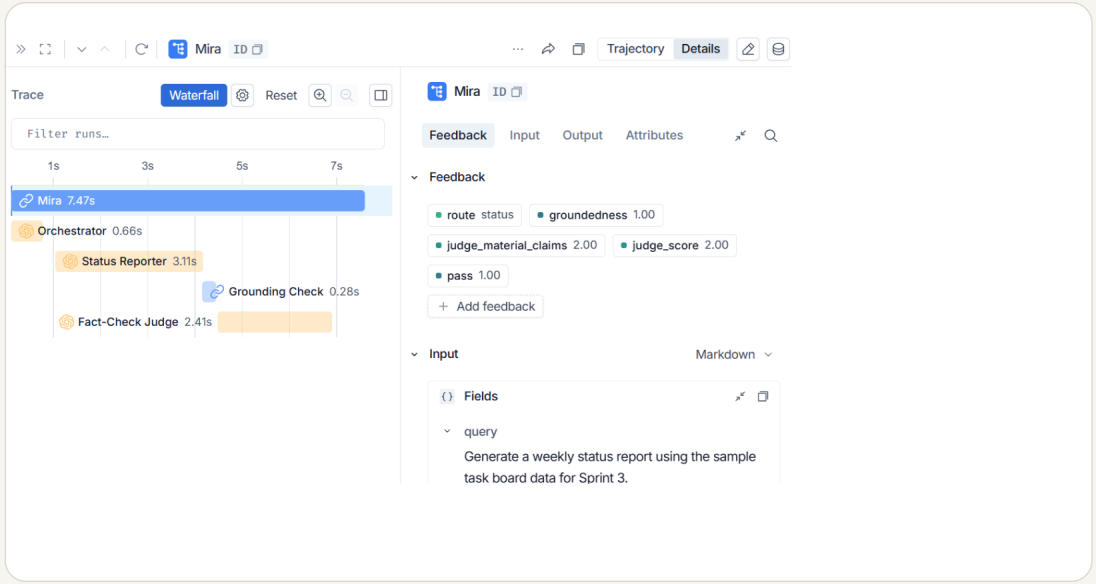
Judge v3 · gpt-4o

material-claim precision; caught real errors, but scored identical claims differently

Decision: the judge is **advisory**. Its score is shown to the PM and logged to LangSmith for review; deterministic checks decide pass or fail.

OBSERVABILITY

Every request traced, scored and compared in LangSmith



Trace: Orchestrator → Status Reporter → checks → judge, with feedback scores



Experiments: each suite run compared on dataset mira-baseline

Per-call model, tokens, latency and cost · scores as trace feedback, live chat included · every test run uploaded as an experiment

COST AND LATENCY

Less than a tenth of a cent per plan

$0.0008

per project plan (3,406 tokens)

$0.009

for all 13 tests (39,216 tokens)

4.4 s

median latency; 12–15 s for plans and risk assessments

≈ $3

per month for 10 PMs (~4,400 requests)

⚠ The gpt-4o judge costs about 8× Mira itself (+$0.07 per suite, ≈ $28/month on every request). Run it on a sample, and skip clarifier and fallback answers.

Tokens measured per run; prices at OpenAI list rates. LangSmith measured the T5 Status Reporter call: 1,546 in + 316 out = $0.0004.

PROGRAM CHARTER

From prototype to 10 PMs in 16 weeks

Done

Prototype

5 capabilities, 13/13 tests, LangSmith live

Weeks 1–4

Harden

Jira/Trello connector, multi-project, chat memory, PII redaction

Weeks 5–8

Pilot

2 PMs on live projects; PM approves every output

Weeks 9–16

Expand and operate

All 10 PMs in waves; Monday alerts; monthly eval review

Success criterion	Target	Measured by
Time to first draft	Plan ≤ 15 min · report ≤ 10 min	PM time logs, before and after
Groundedness	≥ 95% (prototype 100%)	Checks → LangSmith feedback
Material errors in PM review	≤ 1 per 10 outputs	Review sample + judge flags
Adoption	8 of 10 PMs weekly active	n8n executions, traces

REFLECTION AND NEXT STEPS

What the traces taught, and what comes next

Trace findings

The model follows the letter of a rule: it rated 3 of 10 risks and copied prompt examples. Checks catch IDs and numbers, but claim errors like “due soon” for 8-day tasks still slip through.

Improvements

Targeted claim checks in code, a PM approval step that captures corrections, live task-board data, and a judge run on a sample only.

Evaluation → fine-tuning

PM-corrected traces become training pairs. Fine-tune the Risk Assessor only after 100–200 corrections, and only if it beats the prompt on mira-baseline.

Privacy

Task boards hold staff names (UK GDPR). Redact names in traces, use EU hosting, keep short retention, and require PM approval before anything reaches a client.

THE LESSON

Decide what the model is not allowed to do.

Code owns the facts

Schemas own the shape

The model owns the wording

Observability made every failure visible; evaluation made every fix provable. · github.com/agcode-star/MIRA-AI-Project-Assistant